CONSTRUCTOR TECHNOLOGY · EEG MIND READING

Telepathy

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Decoding visual perception
from EEG signals

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ML PIPELINE

Preprocessing → Features → Classification

01	**Preprocessing**	Bandpass 0.5–75 Hz + 50 Hz notch (on top of provided filter & blink removal)
02	**EEG Features**	LaBraM, modified LaBraM (3s), LUNA, CBraMod + engineered features
03	**Video Features**	OpenCLIP, DINOv2, AlexNet embeddings
04	**Classification**	Per-subject RidgeCV: map video features → EEG response, rank by Pearson

CLASSIFICATION

Ridge regression on raw EEG

Per-subject RidgeCV	5-fold CV, alphas = logspace(-3, 6, 10)
Mapping	PCA video features → full EEG response (17 ch × 300 samples)
Selected alphas	1e4 or 1e5 depending on subject
Retrieval	Predict EEG for each of 102 candidates, rank by Pearson correlation
Why it worked	Strong regularization helped generalization; learned models overfit

RESULTS

Test retrieval (102 candidates)

Chance: Top-1 = 0.98% · Top-5 = 4.9%

Subject	Top-1	Top-5	Median Rank
01	5.9%	24.5%	15.0
02	3.9%	24.5%	19.5
03	4.9%	17.7%	20.5
04	4.9%	14.7%	22.5

FEATURE ABLATION

Subject 1 — what worked best

Features	Test Top-1	Test Top-5	Median Rank
AlexNet features.3	3.9%	11.8%	25.5
DINO blocks.6	4.9%	30.4%	13.0
CLIP resblocks.6	3.9%	16.7%	17.0
CLIP resblocks.7	4.9%	21.6%	14.0
Ensemble (submitted)	5.9%	26.5%	14.0

Why Ridge on raw EEG won

CCA	100% train top-1, but test at chance (median rank ~50)
CLIP-style contrastive	Heavy overfitting — memorizes training pairs
MLP regression	Underfits; train already at chance
Ridge (submitted)	Only method clearly above chance on test

KEY TAKEAWAYS

What we learned



- Strong regularization ($\alpha \approx 10^4$–10^5) was critical for generalization
- Ensemble of video features beat any single source on test top-1
- Learned EEG embeddings (LaBraM, LUNA, CBraMod) overfit more easily
- Raw (decimated) EEG + Ridge was more robust than complex models
- DINO features gave the strongest single-model top-5

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Decoding what the mind sees

Thank you!

